

1 Data Preprocessing and Feature Engineering

1.1 Data Source and Initial Challenges

The original dataset was collected from **Ouedkniss**, an Algerian online marketplace. Unlike datasets obtained from structured input forms, Ouedkniss listings are primarily created using *free-text descriptions written by sellers*. Although the scraped dataset appeared structured, most columns were inferred from unstructured text.

The initial dataset contained the following attributes:

- price_preview
- created_at
- city
- spec_Etat
- model_name
- DEDICATED_GPU
- CPU
- RAM_SIZE
- SSD_SIZE
- HDD_SIZE
- SCREEN_SIZE
- SCREEN_FREQUENCY
- SCREEN_RESOLUTION
- RAM_TYPE

In practice, many sellers do not explicitly fill hardware fields. Instead, all specifications are written within a single description block. This resulted in several challenges:

- Inconsistent and informal CPU and GPU naming
- High missingness across multiple columns
- Swapped or incorrect values (e.g., RAM written as SSD)
- Absence of low-level hardware specifications such as core counts or benchmark scores

Since laptop price prediction and clustering are strongly driven by hardware performance, addressing these issues became the core focus of the preprocessing stage.

1.2 CPU and GPU Standardization and Enrichment

1.2.1 Motivation

The CPU and DEDICATED_GPU columns were among the noisiest features. The same processor frequently appeared under multiple informal representations such as “11th i5”, “core i5 11...”, or “intel i5 gen 11”. Furthermore, no direct performance indicators were available.

Initial attempts to treat these fields as categorical variables were quickly abandoned, as they ignored performance similarity and amplified noise. Consequently, we adopted a strategy based on standardization and enrichment using external hardware benchmarks.

1.2.2 PassMark Reference and Specification Datasets

We collected external datasets from **PassMark**, which provided:

- Standardized CPU and GPU names
- Technical specifications such as CPU Mark, GPU G3D Mark, core count, and TDP

These datasets served both as normalization references and as feature enrichment sources for downstream modeling.

1.2.3 Fuzzy Matching for Hardware Mapping

Due to the absence of a fixed naming format, exact string matching was infeasible. We therefore employed **fuzzy string matching** using **RapidFuzz** to map scraped CPU and GPU names to their closest PassMark counterparts.

The mapping process consisted of:

1. String normalization (lowercasing, removal of vendor terms and punctuation)
2. Similarity computation against PassMark entries
3. Selection of the best match
4. Application of a minimum similarity threshold

Entries falling below the threshold were discarded. These were primarily generic mentions such as “core i5”, which do not convey sufficient performance information. The same approach was applied independently to GPUs.

1.3 Handling Integrated GPUs

A large portion of the dataset contained missing values in the `DEDICATED_GPU` column. Manual inspection revealed that this was largely due to laptops relying on integrated GPUs, which sellers rarely mention explicitly.

To address this issue:

- The CPU reference dataset was extended with an integrated GPU name column using domain knowledge
- CPUs were first mapped to PassMark entries
- The corresponding integrated GPU was inferred
- The inferred GPU name was joined with the GPU PassMark dataset to retrieve benchmark specifications

This approach allowed us to recover meaningful GPU information for a substantial number of listings.

1.4 Column Removal Decisions

Several columns were removed after exploratory analysis:

- `city`: exhibited very weak correlation with price and introduced regional noise
- `SCREEN_FREQUENCY`: suffered from extreme missingness
- `created_at`: did not realistically reflect laptop value in this market context

These decisions were guided by both statistical observations and domain reasoning.

1.5 RAM and Storage Processing

Once CPUs and GPUs were standardized, they were used as anchors to clean RAM and storage attributes.

1.5.1 Consistency-Based Imputation

Hardware domain knowledge was used to fill missing values:

- High-end CPUs rarely pair with HDD-only storage
- New-generation CPUs are incompatible with older RAM types such as DDR3

This approach proved more reliable than purely statistical imputation.

1.5.2 Swap Detection and Correction

A frequent scraping error involved swapped RAM and SSD values (e.g., RAM listed as “512GB”). Heuristic rules based on realistic size ranges were used to detect and correct such cases.

1.5.3 Encoding

After cleaning:

- RAM sizes were converted to numeric values (e.g., 8GB $\rightarrow$ 8)
 - RAM types were encoded ordinally to reflect technological progression
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1.6 Screen Feature Processing

Screen-related attributes suffered from inconsistent reporting and missing values.

- Missing values were filled using CPU and GPU performance tiers
- Screen resolutions were encoded as ordinal categories (HD, FHD, QHD, 4K) rather than raw pixel counts

This avoided misleading numeric distances between resolutions.

1.7 Model Name Abstraction

The `model_name` column had high cardinality and was unsuitable for direct encoding. Instead:

- Laptops were grouped by model
- Average CPU Mark and GPU G3D Mark were computed per model
- Models were categorized into low-end, mid-range, and high-end tiers

These tiers were encoded ordinally, and the original model names were dropped.

1.8 Price Cleaning and Correction

1.8.1 Troll Price Removal

Non-serious prices such as sequential values (e.g., 123, 123456) and repeated digits (e.g., 111111) were removed using a predefined list of common troll patterns.

1.8.2 Dinar and Centime Ambiguity

A common issue in Algerian marketplaces is inconsistent use of dinars and centimes. To address this:

- Expected laptop prices were estimated using CPU and GPU component prices
 - Prices outside plausible ranges were corrected using scaling factors
 - Brand multipliers were applied for consistently premium brands (e.g., Apple, ROG)
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1.9 Laptop Condition Encoding

The `spec_Etat` column contained inconsistent textual labels. These were unified into standardized categories and encoded ordinally. Missing values were inferred using indirect indicators such as hardware recency.

1.10 Preprocessing Outcome

The final preprocessing pipeline transformed a noisy, semi-structured dataset into a consistent and enriched dataset suitable for price prediction and clustering. The adopted preprocessing choices were driven by real marketplace constraints, hardware domain knowledge, and iterative refinement rather than purely theoretical assumptions.